

Sec 2.3 – The Derivative Function

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

The following table

gives values of $c(t)$, the concentration ($\mu\text{g}/\text{cm}^3$) of a drug in the bloodstream at time t (min).

t (min)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
$c(t)$ ($\mu\text{g}/\text{cm}^3$)	0.84	0.89	0.94	0.98	1.00	1.00	0.97	0.90	0.79	0.63	0.41

Partner up

Construct a table of estimated values for $c'(t)$, the rate of change of $c(t)$ with respect to time.

We assume the data points are close enough together so that the concentration does not change dramatically between them. Where the change in concentration is increasing, we expect a positive derivative, and when the concentration decreases, we expect a negative derivative.

The data points are equally spaced, 0.1 apart, so $h=0.1$.

The formula to use is:

$$c'(t) \approx \frac{c(t+h) - c(t)}{h}.$$

Which gives us:

$$\begin{aligned} c'(0) &\approx \frac{c(0.1) - c(0)}{0.1} = \frac{0.89 - 0.84}{0.1} = 0.5 \mu\text{g}/\text{cm}^3 / \text{min}. \\ c'(0.1) &\approx \frac{c(0.2) - c(0.1)}{0.1} = \frac{0.94 - 0.89}{0.1} = 0.5 \mu\text{g}/\text{cm}^3 / \text{min}. \end{aligned}$$

Don't forget the units!

t	$c'(t)$
0	0.5
0.1	0.5
0.2	0.4
0.3	0.2
0.4	0.0
0.5	-0.3
0.6	-0.7
0.7	-1.1
0.8	-1.6
0.9	-2.2

How could we improve the estimate? The estimate for the derivative at $t=0.2$ used the interval to the right, it was the average rate of change between $t=0.2$ and $t=0.3$. But we could have used the interval to the left, finding the average rate of change between $t=0.1$ and $t=0.2$.

So, we could average those results for a more accurate answer:

$$c'(0.2) \approx \frac{1}{2} \left(\begin{array}{c} \text{Slope to left} \\ \text{of } 0.2 \end{array} + \begin{array}{c} \text{Slope to right} \\ \text{of } 0.2 \end{array} \right) = \frac{0.5 + 0.4}{2} = 0.45.$$

Let's use the limit of the difference quotient to find the derivative function for a few functions, and we'll see if we can generalize these results.

Partner up

Example: $f'(x)$ where $f(x) = x^2$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2) - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} \\ &= \lim_{h \rightarrow 0} 2x + h = 2x \end{aligned}$$

The last equality follows by continuity and the second to last from the fact that we can cancel since $h \neq 0$.

Example: $g'(x)$ where $g(x) = x^2 + 3x$

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(x+h)^2 + 3(x+h)] - [x^2 + 3x]}{h} \\ &= \lim_{h \rightarrow 0} \frac{[\cancel{x^2} + 2xh + h^2 + \cancel{3x} + 3h] - [\cancel{x^2} + \cancel{3x}]}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 3h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2x + h + 3)}{\cancel{h}} \\ &= \lim_{h \rightarrow 0} 2x + h + 3 = 2x + 3 \end{aligned}$$

The last equality follows by continuity and the second to last from the fact that we can cancel since $h \neq 0$.

$$f(x) = 5$$

$$\text{Solution: } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{5 - 5}{h} = 0$$

We can also find this answer by thinking about the meaning of the derivative - the tangent line to the function $f(x) = 5$ is the function itself and the slope is 0 since the line is horizontal.

This leads to the formula for the derivative of a constant function: **If $f(x) = k$, then $f'(x) = 0$.**

And the derivative for a linear function, since the slope of a straight line is constant the derivative will be constant:

$$\text{If } f(x) = b + mx, \text{ then } f'(x) = \text{Slope} = m.$$

Finally, the derivative of power functions are: **If $f(x) = x^n$ then $f'(x) = nx^{n-1}$.**